Module 06: Learning in Robotics

Learning Objectives

1. Define **Learning** within the context of Robotics.
2. Analyze how Learning interacts with other components of the Active Inference framework.
3. Apply specific constraints of Robotics to the formal definition of Learning.

Introduction

This module explores **Learning**. In the **Robotics** curriculum, we approach this topic with a focus on specific applications and theoretical depth appropriate for the audience. Learning is a critical component of the 8-part Active Inference spine, bridging the gap between [Previous Topic] and [Next Topic].

Key Concepts

1. Learning as a Markov Blanket Boundary

How does Learning define the boundary between the agent and the environment?

2. Generative Models of Learning

What parameters involved in Learning must be optimized to minimize variational free energy?

3. Active Inference Dynamics

How does the process of Learning drive the perception-action loop?

Applications

In Robotics, we see Learning manifest in: * **Specific Example 1**: [Add domain-specific example here] * **Specific Example 2**: [Add domain-specific example here]

Conclusion

Understanding Learning allows us to better model complex adaptive systems. In the next module, we will expand on this foundation.